Dept. of Computer Engineering, NIT Kurukshetra

ITPC-205 (OOP using JAVA), Final Written Exam

Instructions: Adhere to the word limit. **Answer all subparts of a question at the same place.**

All questions are compulsory.

1. Write a program that prompts the user to read two integers and display the product of all digits. If the products are equal then the input is correct else it should throw a custom exception *InputMismatchException* to print 'wrong input'. Your program should prompt the user to read the number again if the input is incorrect [7]

2. Write a Java program to create an interface Sortable with a method sort (int[] array) that sorts an array of integers in descending order. Create two classes, QuickSort and MergeSort that implement the Sortable interface and provide their respective implementations of the sort() method. [10]

Sample run:

Quick Sort (Descending Order): 9 8 6 5 3 2 1 0

Merge Sort (Descending Order): 9 8 6 5 3 2 1 0

3. Create a custom package and demonstrate how to make specific classes and members within that package accessible or hidden from other packages using access modifiers. [7]

4. a. Give the diagram cycle of a thread and explain each state in not more than two lines. [4]

   b. Give 3 key differences between interface and abstract class. Explain each. [3]

   c. Define polymorphism. Explain two ways to achieve polymorphism. Give 3 key differences between the two ways to achieve polymorphism. [1+2+2]

   d. Give 4 key differences between throw and throws. [2]

5. Predict the output with a reason in 2 lines. [6*2=12]

i)

```
public class Test
{ public static void main(String[] args)
{
System.out.println("Hi, ABC, good".matches("ABC "));
System.out.println("Hi, ABC, good".matches(".*ABC.*"));
System.out.println("A,B;C".replaceAll(",,","#"));
System.out.println("A,B;C".replaceAll("[,;]","#"));
String[] tokens = "A,B;C".split("[,;]");
for (int i = 0; i < tokens.length; i++)
System.out.print(tokens[i] + " ");
}}
```

ii)

```
public class Test {
public static void main(String[] args)
{ String s = "Hi, Good Morning";
System.out.println(m(s));
}

Public static int m(String s)
{ int count = 0;
for (int i = 0; i < s.length(); i++)
if (Character.isUpperCase(s.charAt(i)))
count++;
return count;
}}
```

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iii)

```
public class Test
 { public static void main(String[] args)
 { A a = new A(3);
 } }

class A extends B { public A(int t)
{System.out.println("A's constructor is invoked");
}}

class B { public B()
 { System.out.println("B's constructor is invoked"); } }
```

iv)

```
public class Test
 { public static void main(String[] args)
 { Object circle1 = new Circle();
Object circle2 = new Circle();
System.out.println(circle1.equals(circle2));
} }
class Circle
{ double radius;
 public boolean equals(Object circle) {
 return this.radius == ((Circle)circle).radius;
} }
```

v)

```
public class Test
 { public static void main(String[] args)
 { Object circle1 = new Circle();
Object circle2 = new Circle();
System.out.println(circle1.equals(circle2));
} }
class Circle {
 double radius;
public boolean equals(Circle circle)
 { return this.radius == circle.radius; } }
```

vi)

```
public class Test
 { public static void main(String[] args)
 { new A();
new B(); }
}

class A {
 int i = 7;
public A()
 { setI(20);
System.out.println("i from A is " + i);}

 public void setI(int i)
 { this.i = 2 * i; } }

class B extends A {
public B()
 { System.out.println("i from B is " + i);
}

 public void setI(int i)
 { this.i = 3 * i; } }
```